CycleTwin

Cycle-by-cycle late or absent ovulation research

Team CycleTwin / VIT

A fitted synthetic forecasting workflow with explicit safety gates

Meet Kushi: fictional persona, never performance evidence

Synthetic research prototype. Not for diagnosis, contraception or conception planning.

The research problem

Monitoring irregular cycles without claiming a PMOS diagnosis

Target: generated ovulation after the configured cutoff, or generated absence

India context: 7.2% by NIH criteria and 19.6% by Rotterdam criteria

Published prevalence depends on definition. It is not the simulator mixture or model accuracy.

Meet Kushi

Fictional synthetic persona - not a real patient - not performance evidence.

21 years, year 3 B.Tech

Hostel life, exam-season sleep debt, mother with type 2 diabetes, smartwatch wearer.

Actual generated past lengths: 41, 38, 41, 38, 44, 51 days

Selection uses generator truth and a declared seed, never predicted risk

Static and dynamic input streams

EHR-lite is implemented and explicitly oracle-adjacent

Static: generated IR latent, androgen, age, BMI and family history

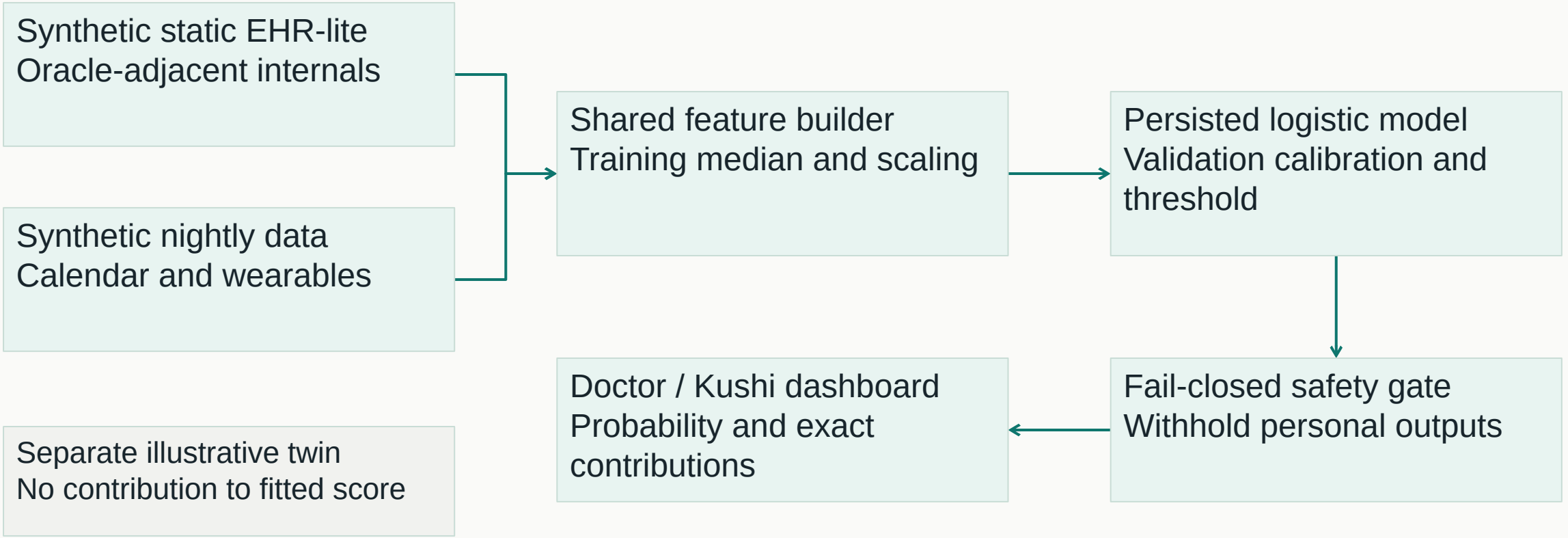
Dynamic: calendar history, RHR, HRV, steps, sleep and temperature

No measured hormone values. The XML parser demonstrates input shape only.

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One fitted prediction path

The dashboard loads the same persisted pipeline used by the benchmark



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Training and selection

Every explicit operating choice uses train or validation data

Patient counts per development seed: train 840, validation 180, test 180

Training medians and scaling, logistic regression, validation Platt calibration

Validation Brier-gain rule selects richer features or calendar fallback

Validation F1 chooses the threshold. Test outcomes never choose that threshold.

Synthetic development evidence: day 10

Seed means describe the current generator, not clinical forecasting

Day	Tier	Mean AUROC	Mean Brier	Mean day MAE
10	0	0.838	0.163	5.004
10	1	0.838	0.163	5.004
10	2	0.838	0.163	5.004
10	3	0.838	0.163	5.004

Synthetic development evidence: day 14

Seed means describe the current generator, not clinical forecasting

Day	Tier	Mean AUROC	Mean Brier	Mean day MAE
14	0	0.838	0.163	5.132
14	1	0.838	0.163	5.132
14	2	0.850	0.157	6.069
14	3	0.843	0.161	5.565

Synthetic development evidence: day 18

Seed means describe the current generator, not clinical forecasting

Day	Tier	Mean AUROC	Mean Brier	Mean day MAE
18	0	0.838	0.163	5.439
18	1	0.839	0.163	5.439
18	2	0.906	0.119	5.267
18	3	0.907	0.119	5.267

Frozen synthetic evidence: day 10

Reserved seeds, unchanged train/validation choices, one guarded run

Day	Tier	Mean AUROC	Mean Brier	Mean day MAE
10	0	0.839	0.164	5.142
10	1	0.839	0.164	5.142
10	2	0.839	0.164	5.142
10	3	0.839	0.164	5.142

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Frozen synthetic evidence: day 14

Reserved seeds, unchanged train/validation choices, one guarded run

Day	Tier	Mean AUROC	Mean Brier	Mean day MAE
14	0	0.839	0.164	5.251
14	1	0.839	0.164	5.251
14	2	0.855	0.155	6.790
14	3	0.850	0.158	6.319

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Frozen synthetic evidence: day 18

Reserved seeds, unchanged train/validation choices, one guarded run

Day	Tier	Mean AUROC	Mean Brier	Mean day MAE
18	0	0.839	0.164	5.570
18	1	0.839	0.164	5.570
18	2	0.907	0.117	5.449
18	3	0.907	0.117	5.449

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Evidence at the latest sensor snapshot

Predeclared temperature sensor tier, descriptive seed means

Synthetic day 18, Tier 2: AUROC 0.907, Brier 0.117

3,240 synthetic test cycles across 3 frozen seeds

A separate frozen evaluation artifact is available.

Standard average precision and calibration diagnostics are available from stored predictions.

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Frozen sequential alert-policy evidence

Earliest safe threshold crossing only; all values are synthetic

Tier 2: 1,634 alerts across 3,240 generated cycles; alert precision 0.730

Missed generated events: 484/1,677 (0.289); false alerts: 1.484 per synthetic patient-year

Median generated-event lead time: 17.000 days among 790 true-positive cycles with a generated ovulation day

Safety is evaluated at every snapshot; no later score can change an earlier alert.

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Frozen synthetic robustness extension

Fixed Tier-2 day-18 model and threshold under predeclared perturbations

At 60% additional forced missing temperature readings: AUROC 0.884, Brier 0.140, safety withholding 1.000

At +0.01°C/day drift: AUROC 0.906, Brier 0.300

At a +4-day generated timing shift on independent synthetic cohorts: AUROC 0.863, Brier 0.145

These are simulator sensitivity tests, not device robustness or clinical validation.

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Timing-posterior check

Train-only calendar/static prior plus temperature, RHR and HRV likelihoods

Synthetic Bayesian-posterior Brier 0.098; fitted logistic Brier 0.117

Twin minus logistic Brier -0.018, patient-cluster 95% CI [-0.022, -0.014]

noninferior synthetic check; still separate interpretation layer

Separate from the logistic score, threshold, safety gate and alert policy.

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Simulated assay-like extension

Post-freeze, noisy simulator proxies; neither clinical labs nor measured hormones

Tier 2 plus simulated assay-like indices: AUROC 0.907, standard AP 0.885, Brier 0.118

Original Tier-3 oracle-adjacent internal reference: AUROC 0.907, Brier 0.117

The proxy uses positive exp(latent) values and independent 15% CV assay noise.

It retains simulator-internal leakage risk and is not a clinical EHR-fusion claim.

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Local operations measurement

One persisted frozen Tier-2 model on the build host

Cold new-process first prediction: 1193.724 ms

Warm inference: mean 0.533 ms, p95 0.565 ms across 1,000 calls

Local process peak RSS: 193104 KiB

Host-specific engineering measurement, not a production-service SLA.

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Frozen synthetic conformal coverage

Marginal classification sets and separate heuristic timing bands

Day-18 Tier-2 classification-set coverage: 0.901; mean set size 1.155

Heuristic ovulation-day band coverage: 0.056; mean width 24.000 days

The classification-set result is near its synthetic 90% target.

The day-band result is poor; do not use it as a reliable timing interval.

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What the clinician view shows

Decision-support research display

Fitted probability beside its own validation threshold and held-out model evidence

Exact additive feature contributions and explicit imputation

Heuristic day estimate and conditional, illustrative day band

Data-quality ribbon and the reason for any withholding

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Guardrails live

Missing safety information causes abstention

Contraception, thyroid and prolactin flags, age and pregnancy-related states

Early post-menarche, perimenopause and insufficient observed temperature

Withholding removes personal scores, intervals, explanations, sandbox outputs and alerts

The workflow is synthetic and its safety effectiveness has not been clinically evaluated

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The twin and assumption sandbox

Clear boundaries around simulated physiology

Reproductive and metabolic equations generate an illustrative state trajectory

The twin does not affect the logistic score or its reported performance

The sandbox edits invented model inputs. It does not predict a treatment effect

Glucose threshold crossings are simulated counts, not an evaluated glucose forecast

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Where it fails

All declared Kushi scenarios remain visible

Display the fixed scenario seeds with misses, false alarms or withholding unchanged

Different snapshots can fall back to calendar even when sensors are available

A class set containing both outcomes does not identify one result

Robustness, state-estimator and conformal coverage evidence remain separate work

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Validity limits

Synthetic recovery is not prospective evidence in women

Labels and sensor changes share generator assumptions

EHR-lite includes a label-generating internal latent

Old test seeds were reused during development

The frozen synthetic sequential alert policy is evaluated; robustness and clinical safety effectiveness remain separate work

Reproducible demonstration

Public source, inspectable artifacts and explicit next work

README, model cards and the audit link every claim to evidence

Persisted inference reproduces the benchmark and exposes exact feature calculations

The required video remains a human recording and upload task

Clinical validation, clinician feedback and sim-to-real evaluation remain outstanding

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